

Chapter 16

Data: Access, Generation, Use



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16.1 Introduction

Data can be a confusing topic. The breadth and variety of data and related issues posed by a potential university–industry (UI) collaboration are daunting. Deciding what data is needed, how to generate or access it, if and how to protect it, how and by whom it can be used are important to discuss prior to the onset of any collaboration or research project. Technical issues such as the format of the data, how to access, transfer, or exchange it, how to curate it, how long to keep it, and how and whether to return or destroy it are often part of a proposed project discussion.

Data comes with challenges as well as opportunities. Quality and provenance of data become important, particularly when data is being used for decision-making about healthcare, policy, regulation, security, and training of models and software that support these and other key objectives. The ability to aggregate datasets and compare data across datasets may depend on the development of common ontologies or the ability to recognize similar words as meaning the same thing, for example, cardiac and heart, breathing and respiration. The growing trend to enlist the use of artificial intelligence (AI) to recognize and resolve this issue, while encouraging, requires access to and use of competent, reliable data.

Federal, state, and international regulation and legal status of data is changing daily and has direct effects on data issues in UI collaborations. How data privacy and security laws affect access to, exchange, and use of some types of data may affect collaborators' ability to realize their respective objectives. Data-sharing policies of US funding agencies, nonprofit sponsors, and universities assume there is benefit to broad sharing of data for research, decision-making, and policy

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development. However, questions remain: how is the quality of data ensured and monitored; how is its potential commercial value assessed and, if necessary, protected? This chapter is intended to alert readers to these and other potential issues and provide advice and strategies for dealing with them.

16.2 Background

For background, this section begins by exploring the meaning of “data” in UI agreements and provides context as to the types of agreements that are formed in UI interactions involving the use and sharing of data.

16.2.1 Data Definitions

Perhaps the most challenging feature of “data” in UI relationships is that there is no universal definition. A generic definition can be found in any dictionary but even there the term has multiple meanings. For example:

‘data’ 1. factual information (such as measurements or statistics) used as a basis for reasoning, discussion, or 2. information in digital form that can be transmitted or processed, or 3. information output by a sensing device or organ that includes both useful and irrelevant or redundant information and must be processed to be meaningful. (Merriam-Webster Dictionary, [2024](#))

With respect to UI relationships, the University Industry Demonstration Partnership’s (UIDP) “Contract Accord 14: DATA” defines data broadly:

“Data” means a set of recorded information that is provided by one party to another party for use under defined conditions. (UIDP, [2020](#))

The concept of “recorded information” appears in the general policies of federal research sponsors in lieu of the term “data.” For example, the general Federal Acquisition Regulations define data as “recorded information, regardless of form or the media on which it may be recorded.” The regulation goes on to state that recorded information includes computer software and software documentation but does not include incidental information, such as financial, administrative, cost or pricing, or management information. (Acquisition.Gov, [2024](#)).

Individual federal agencies’ data definitions reflect their policies about how data is commonly generated or shared among parties, the agency, and the public. For instance, the National Institutes of Health (NIH) defines “data” as “recorded information, regardless of the form or media on which it may be recorded,” and offers a long list of data registries (NIH, [2024a](#), [2024b](#)). The US Department of Defense defines a subset of information called “technical data” as “recorded information, regardless of the form or method of the recording, of a scientific or technical nature.” Software documentation is included as technical data, but the software itself is not

(Acquisition.gov, 2023). The US Department of Education defines “Education records” that meet certain criteria; “(1) directly related to a student; and (2) maintained by an educational agency or institution or by a party acting for the agency or institution.” Education records are defined and regulated under the Federal Educational Rights and Privacy Act, commonly known as FERPA (Federal Education Rights and Privacy Act of 1974; US DOE, 2024; US Congress, 1974). Like federal research sponsors, industry sponsors also incorporate definitions of data that are specific to the work conducted in their sponsored research agreements.

16.2.2 Data-Sharing Terms in UI Agreements

Data-sharing provisions are included in various UI agreements that are not governed by federal regulations. These agreements include Sponsored Research Agreements (SRAs), Materials Transfer Agreements (MTAs), Data Use or Transfer Agreements (DUAs, DTUAs), Confidential Data Agreements, Non-Disclosure Agreements, or Partnership Intermediary Agreements (CDAs, NDAs, PIAs), option and license agreements, Clinical Trial Agreements (CTAs), and Services Agreements (SAs). Each of these agreements would ideally include a context-specific definition of data but it does not take much review of these agreements to see that is not always the case.

In these various types of agreements, data is often lumped into broad, often circuitous, definitions of legally protectable types of intellectual property, such as inventions or copyrights or included in other ambiguous definitions that refer generically to “information.” Here are some examples that are not precise enough to determine what data is covered:

- “*Intellectual Property*” can encompass inventions, ideas, concepts, discoveries, developments, know-how, patent rights, trade secrets, techniques, methodologies, modifications, innovations, improvements, derivatives, writings, documentation, *data*, and related rights.
- “*Inventions*” may include *information* in addition to inventions, or discoveries (whether patentable, copyrightable, or not), innovations, suggestions, ideas, communications, and reports, conceived, or reduced to practice.
- “*Results*” may represent the results of a project, including test results, *data*, technical information, procedures, and reports.
- “*Confidential Information*” may mean all valuable *information* acquired, or developed in a project or used by the sponsor for a competitive advantage over those who do not know such information.

Clauses such as these create drafting chaos since they require clarification of the applicability of the definitions to other terms, such as confidentiality, rights and licenses, publication, privacy, and liability and are subject to different interpretations by the parties. Conflicts can later arise if differing interpretations are not exposed during negotiation.

Data deserves and requires treatment as a discrete input to, or result of, a UI interaction. Data requires clear definition, articulated permitted scope of use, and appropriate limitations on disclosure, all of which apply specifically to the context and circumstance to which the data applies. The following simplified example segregates and defines data:

“*Data*” means a set of recorded information that is generated in performance of a project and contained in reports required to be provided to the sponsor as described in the statement of work.

“*Rights in Data*” a sponsor may use the data in any way and for any lawful purpose consistent with the agreement.

This example, however, may require the addition of other clauses that contain limitations on use of the data, such as a privacy clause covering Personally Identifiable Information (PII).

It is a best practice for a statement of work to explain and exclude, if necessary, data used or acquired by the university in the performance of the project that is subject to limitations or third-party restrictions on the sponsor’s rights to use the data.

16.2.3 Data Ownership Versus Use

Ownership or control of single data points is not generally an issue in UI interactions for data resulting from the conduct of research, job-related tasks, and similar activities. Discussions in these cases are about the compiled, reported, aggregate data that results from the activity and are provided to the industry collaborator or sponsor in a report or as a dataset exchanged between the parties. However, data resulting from other kinds of activity that involves human subjects requires special consideration. Examples include clinical trials and student projects, both of which depend on data from particular individuals and require some sort of approval or authorization from or on behalf of the individuals who are the subjects of the data.

In sponsor-initiated clinical trials, an industry sponsor often requests not only ownership of resulting clinical data but also restrictions on use by the sites that are enrolling the subjects. Considering that subject data is generally entered into the subjects’ medical records as well as into case report forms or an electronic database, the data can be reproduced and is not uniquely available to the sponsor. Also, simply stated, “The data obtained in a clinical trial are facts and cannot formally be owned by anybody” (van der Beek & Cohen, 2023). In these cases, data should be clearly defined, and terms discussing ownership of the data should be avoided in favor of terms describing the respective parties’ rights to use the data. When a university is a clinical trial site, these uses would include publication, research, education, and patient care. For the industry sponsor, the uses would include any purpose allowed by the informed consent of the subjects (e.g., research, development, regulatory

filings) but may not allow access to identifiable patient data, except for monitoring or auditing performance of the study.

Ownership of data as discrete facts is not protected by copyright law. However, databases, while not expressly listed as copyrightable subject matter under US copyright law, “may be considered copyrightable as a form of compilation” if “selected, coordinated, or arranged in such a way that the resulting work as a whole constitutes an original work of authorship” (US Copyright Office, 1992). Databases may be compiled by a particular entity in such a way that they cannot be efficiently reproduced. These databases may be publicly accessible on websites that often require login access that binds the user to terms of use described on the website when the user accesses the data. The owners of these websites retain the copyright to the database, unless they expressly indicate that they dedicate the database to the public, that is, place it in the public domain, thus abandoning their copyrights. There is a critical distinction between publicly accessible databases that are subject to terms of use and those that are in the public domain and available for unrestricted use.

Ownership of databases that are generated by contributions submitted from various sources is more akin to a stewardship role since the provenance of the individual data points once submitted may be divorced from their source or otherwise difficult to ascertain. Also, the database stewards who control the curation and management of the database may change the formatting, edit the data points to have a common ontology, and develop policies and procedures for monitoring auditing, permitting access, use, and reference to make the database a useful tool.

Terms in a UI agreement, under which a database is to be created by a university and that cannot be cost-effectively re-created, may recognize the implicit value of the database. Terms can be tailored to the needs and intentions of the parties. For instance, restricted access to the entire database may be given to the industry sponsor for some period of time. As another example, a semi-exclusive license may be given to the industry sponsor, offering exclusive commercial rights while allowing the university to publish about and use the database, as well as to make the database available to other researchers for research and development but not allowing disclosure to for-profit entities.

16.3 Options, Solutions, and Impact

Universities and companies have developed policies and standards for managing, sharing, and protecting data, aimed at complying with laws and regulations while enabling relationships for their mutual benefit. Policies and standards help universities and industry resolve key issues, such as data ownership, use, sharing, security, privacy, retention, and governance. Some examples are provided in this section.

16.3.1 *University Data Policies*

Ivan Novikov, CEO of Wallarm, recommends that organizations implement a data governance framework that

defines policies, procedures, and tools to manage the entire data lifecycle, from creation to disposal. This framework can involve different stakeholders and domains, such as business, IT, legal, and ethics, and can cover various aspects, such as data quality, security, privacy, compliance, and ethics. (Novikov, [2024](#))

Frameworks are often expressed by universities in data-related policies that apply to employees of entities that engage in activities that use or generate data. Some policies may address data specifically while others treat data as intellectual property, trade secrets, or proprietary information.

Many policies classify data into categories related to the required levels of security or privacy and describe procedures corresponding to the classifications. These classifications lead some university policies to recognize that research data needs special consideration. The University of Wisconsin-Madison has a comprehensive policy that broadly defines “Institutional data” as follows:

Information, regardless of medium, generated, collected, stored, maintained, transmitted, enhanced, or recorded by or for the university to conduct university business. It includes data used for planning, managing, operating, controlling, or auditing university functions, operations, and mission. It does not include data generated by research, or data owned or generated by a party outside the university when used in research, conducted at the university, under the auspices of the university, or with university resources. (University of Wisconsin-Madison, [2024b](#))

The policy cascades to other referenced policies. For instance, the Data Classification Policy breaks “Sensitive Information” into four levels and lists data types, borrowing from work done by Michigan State University. Level 2 includes “Institutional data” whose public disclosure is restricted by a contract that contains a confidentiality obligation, while Level 4 is described as

Institutional data whose value would be lost or reduced by disclosure in advance of the time prescribed for its authorized public release, or whose disclosure would otherwise adversely affect the university financially, including but not limited to,

- Research data or results prior to publication or the filing of a patent application
- Non-patentable technical information or know-how that enhances the value of a patented invention or that has independent commercial value.

(University of Wisconsin-Madison, [2024a](#))

Under Wisconsin’s classification policy, the sensitivity of institutional data in Level 4 relates to the timing of publications and disclosures of information that could affect patent filing. The University of Washington has a policy specifically related to “Research Data,” which is defined as

information, records, and tangible products arising from or associated with research conducted at, under the auspices of, or using the resources of the University. Research Data includes both intangibles (e.g., information and copyrighted works such as software and expressions of information) and tangibles (e.g., cell lines, biological samples collected for

research purposes, synthetic compounds, organisms, and originals or copies of laboratory notebooks).

The university then claims ownership of all research data “except as otherwise provided by an agreement with a third party, a law, or University policy, such as copyright policy.” The policy goes on to describe the principal investigator’s stewardship responsibilities, preservation requirements, and procedures for transfer of research data from the university (University of Washington, 2008).

16.3.2 *Data-Sharing Benefits, Risks, and Attitudes*

Data is a frequently discussed topic in universities, industry, and governmental agencies, and is often a critical topic in UI interaction but data sharing has not been accepted by all researchers across scientific disciplines as the preferred way to share knowledge. Efforts to embrace data sharing have described various benefits for both researchers and research sponsors, including better quality of data, stimulated interest, elevated recognition and prominence of researchers and entities, more collaboration, pursuit of unique rather than duplicated research, and data-driven decision-making (USGS, 2024). Researchers can assess the validity and reproducibility of published research results if they have access to the research data and research methodologies that produced it.

A focus group study on data sharing and research data management with researchers from five disciplines (atmospheric and earth science, computer science, chemistry, ecology, and neuroscience), found that, despite the clear benefits of sharing, “many scientists do not share their research data” (Donaldson & Koepke, 2022). Another study recognized data sharing as one of the “cornerstones of modern science that enables large-scale analyses and reproducibility” (Tedersoo et al., 2021). This study “evaluated data availability in research articles across nine disciplines in *Nature* and *Science* magazines and recorded corresponding authors’ concerns, requests and reasons for declining data sharing.” The paper summarized and cited related studies and found that most scientists favor data sharing but also worry about priority in publishing and potential commercial use of their results without their input or consent (Tenopir et al., 2015; Stuart et al., 2018). This concern, as well as the terms contained in UI agreements, may limit the interest in, as well as the ability to share or publish, data resulting from a sponsored project.

Other studies indicate that data availability increases a paper’s citation rate, which should be considered a benefit for university scientists (Tedersoo et al., 2021). But studies also found that “much of the published data is still essentially unavailable for integration into secondary data analysis and evaluation of reproducibility” (Begley & Ioannidis, 2015; Open Science Collaboration, 2015; Hardwicke & Ioannidis, 2018). Tedersoo goes on to analyze data sharing across disciplines from papers published in *Nature* and *Science* over a 20-year interval (2000–2019). Less than 40% of the *Nature* and *Science* paper authors granted data requests from

Tedersoo. Many reasons for declining data requests were cited, including lack of time to search for the data, loss of data, and privacy or legal concerns.

Corporate data sharing poses different concerns than academic or governmental data sharing. One study explored the legal requirements and business incentives to provide access to extremely large and complex datasets (i.e., “Big Data”) from private corporations for statistical analysis and research aimed at finding solutions to public problems. Some of the barriers to private companies include concerns about losing their competitive advantage; legal, privacy, and confidentiality obligations covering client information that may be exposed in the data; and practical limitations such as the costs of the necessary infrastructure and training staff for what the companies may see as unrelated to their core business activity. Perhaps the greater limitation, like academic data, is a false presumption that the necessary data is readily available for sharing (Klein & Verhulst, 2017).

In addition to privacy and confidentiality concerns, there are security and regulatory compliance requirements and complications with companies sharing data across jurisdictions. Big Data that comes from multiple sources, including social media platforms, retailers, financial services companies, and governmental entities, presents issues around data ownership and control, as discussed above. More generic concerns about using Big Data for policy development and public problem solving are how to account for bias in the data obtained from the various sources (unless studies are designed to account for potential bias). Example challenges include variations among developing countries, urban/rural environments, socioeconomic status, and education levels (Klein & Verhulst, 2017).

Data sharing by or among companies and others may be influenced by competition in the company’s market and whether the sharing is between companies that are competitors versus those whose products are horizontally differentiated. These concerns are also present in data sharing from industry sponsors to universities, which require discussion of contact terms that may expressly or indirectly restrict the university’s right to publish research results derived from the data. Sponsors will naturally be concerned about publishing data that may be useful to their competitors or that may otherwise adversely affect their business.

Klein and Verhulst identify six types of incentives or motivations for companies to make data accessible, offering motivation for university data sharing:

- Reciprocity—sharing for mutual benefit, or to reciprocate individual or societal sharing
- Research—problem solving, insights, identification of new talent, development of business models
- Reputation—enhance corporate image to attract customers
- Revenue—generate income from sale or license to use data
- Regulatory compliance—e.g., employment data (race, ethnicity, gender)
- Responsibility and philanthropy—value from socially responsible behavior or advancing causes, e.g., improving education

16.3.3 Data Governance: Privacy

Access to large amounts of data has been enabled by various curation, storage, and retrieval mechanisms. As access has increased, concerns have grown for the protection of privacy and security, along with how sensitive data is obtained, accessed, and used.

Societal focus on privacy and control of one's personal information has and continues to spur the growth of laws requiring protection of such information and restrictions on use, storage, and sharing. Meanwhile, the commercial and scholarly need for this data to develop products and processes, as well as to understand the epidemiology of disease and promotion of health, have increased. While society seeks to establish norms and translate them into laws and regulations, companies and universities struggle with the age-old dilemma between what can be done and what should be done, as well as how to facilitate a culture that understands generation and use of data and embraces data access and sharing.

Federal and state laws protecting privacy generally cover data that is personally identifiable. HIPAA (Health Insurance Portability and Accountability Act of 1996, US Congress, 1996) uses data elements to define Protected Health Information. FERPA has a relatively short list of items that constitute PII but also defines and excludes types of records that are covered as Education Records. HIPAA established the first set of national standards for the protection of certain health information. The primary goal of HIPAA is to allow the free flow of information for personal and public health while protecting an individual's health information (DHHS, 2023). The goal of FERPA is to protect the privacy of students and their parents (US DOE, 2014). Recognizing the complexity, scope, and potential benefits of sharing education records, the Department of Education created the Privacy Technical Assistance Center (PTAC) to assist state and local educational agencies and educational institutions with determining the conditions that allow the disclosure of PII from Education records to third parties, such as researchers, contractors, volunteers, and journalists. PTAC provides a checklist summarizing the requirements for written agreement under the studies exception to FERPA.

The Privacy Act of 1974 is focused on the privacy of data collected by the government and maintained by a federal agency, such as the Census Bureau. The Privacy Act was enacted to address concerns about the abuses presented by the government's increasing use of computers to store and retrieve personal data by means of a universal identifier—such as an individual's Social Security number. These concerns were exposed in a Congressional advisory committee report “that was the first comprehensive study of the risks to privacy presented by the increasingly widespread use of electronic information technologies by organizations, replacing traditional paper-based systems of creation, storage, and retrieval of information” (HEW, 1973).

The Privacy Act allows most agencies to promulgate rules but does not allow use, sharing, or dissemination of data in identifiable form, though it does not define what is considered “identifiable form.” The Congressional briefing of December 2023

(CRS, 2023) looks at the Privacy Act in light of developments in computer science and methodologies used as Privacy-Enhancing Technologies (PETS, US DOJ, 2020). The report seems to acknowledge that restrictions in the Privacy Act limit research use: “While government information can be inherently valuable for researchers, members of the public, and other agencies or governments, uncontrolled access to information may also put individual privacy at risk. Under the principle of purpose specification and use limitation, agencies are to balance the utility of the information against threats to privacy by providing notice of the specific purpose and use of individually identifying information” (CRS, 2023). In response to a suggestion in the report that a national “data warehouse” could “provide access to agencies and researchers to foster learning, improve programs, and reduce the cost of studies,” the Senate Committee on Operations felt the risk to privacy and confidentiality of a national data bank would be too great without statutory guarantees (CRS, 2023).

Prior to a December 2023 briefing, the Office of Management and Budget (OMB) released a memorandum “providing for the establishment of a standard application process through which federal agencies, intergovernmental organizations, and researchers may apply to access confidential data assets” using authority granted to OMB under the Confidential Information Protection and Statistical Efficiency Act (OMB, 2022). The memorandum establishes a standard application process that federal agencies, tribal and territorial governments, researchers, and other individuals may use to apply for access to confidential data sets for “statistical inquiries, from basic research to applied analysis that is highly program specific” (OMB, 2022).

In addition to these federal laws, many states have adopted separate consumer privacy laws. These laws are not uniform and the array of them is changing fast. As of this writing, there are 20 states with data privacy laws in place and several more pending in other states (Bloomberg Law, 2024). The state laws range from narrow to comprehensive and differ in how they are to be implemented and enforced. Most of these laws apply across a range of industries but may not apply to nonprofit organizations and certain types of data.

The European Union enacted the General Data Privacy Regulation (GDPR, EU, 2016). The length and complexity of GDPR and the fact that EU Member States can enact their own implementing regulations make the GDPR hard to interpret and apply to specific situations involving data transfers to countries outside of the EU. A blogger reviewing recent cases with conflicted outcomes explains the problem, “The EU seems ambivalent about deciding how far it can expect third countries to adopt data protection standards similar to its own. Moreover, Data Protection Authorities (DPAs) often fail to scrutinize data transfers to third countries that may lack the rule of law. ...It is essential for the EU to elevate the public discourse so that the global significance of data transfers is recognized” (Kuner, 2024).

The European Data Protection Board issued a response to the request from the European Commission for clarifications on the consistent application of the GDPR focusing on health research (EDPB, 2021). Their response indicated that “the EDPB will provide further clarification on the requirement of a legal basis for further processing for scientific research purposes by the original or a subsequent controller,

also taking into account Recital 50 and Article 6(4) GDPR, in its Guidelines on processing personal data for scientific research purposes (presently under preparation, due in 2021).” To date, this clarification has not been provided.

Note that the EU recently adopted the Data Governance Act “to make more data available and facilitate data sharing across sectors and EU countries in order to leverage the potential of data for the benefit of European citizens and businesses” (EU, 2022). Other countries have also enacted data privacy laws similar to the GDPR. The International Association of Privacy Professionals (IAPP) tracks both US and global privacy laws and makes these directories available online through its Resource Center though access for some content is for members only (IAPP, 2024). Some law firms and companies that provide data privacy services post occasional updates online (e.g., Husain & Enzuzo, 2024).

16.3.4 Data Governance: Security

The proliferation of data and the need for and in some cases technological ease of data capture and sharing has prompted a corresponding increase in data security laws and regulations. Data security involves not only different types of data that have varying needs for restricted access but also the systems for storing and transmitting such data both within an organization and between organizations. As with privacy regulations, companies that provide data security advice, and solutions often offer good summaries on their websites. For instance, Fortinet’s home page offers sections on data security definitions, the importance of data security, benefits of data security, the difference between security and privacy and best practice for securing both, types of data security, risks to security, security solutions, and data security regulations (Fortinet, 2024). Many of these sites address inward-facing issues but do not directly discuss how data security relates to data sharing.

Academic papers and other published sources covering data sharing and security are largely focused on electronic cloud-based data, or data gathered via smart devices. This is particularly true for health data obtained from individuals via apps. The latter offers great benefits through ease of use and speed of data transmission. One author cautions: “By using smart devices for almost any purpose, contact information, calendar entries, or even banking information can be found on these devices. Due to the large volume of sensitive data generated and stored on smart devices, these devices are a prime target for hackers and data thieves” (Stach & Mitschang, 2016). This article then explains and offers the secure data container as a solution for web data sharing with information security.

In the United States, the National Institute of Standards and Technology (NIST) has long been regarded as the lead agency for data security standards. Through its National Cybersecurity Center of Excellence (NCCoE), NIST works with industry experts and technology vendors to address data security challenges and to develop practical, standards-based guidance for organizations to implement to meet their security and privacy needs (NIST 2012). The cybersecurity framework created by

the NCCoE is voluntary. NIST is also responsible for creating mandatory standards to protect and secure government information. NIST SP 800-53 (NIST, 2020) “provides a catalog of security and privacy controls for information systems and organizations to protect organizational operations and assets, individuals, other organizations, and the Nation from a diverse set of threats and risks.”

Since all federal government agencies must comply with these NIST standards, they are often cited in federally funded research contracts for projects conducted by companies or universities. The standards are reflected in several Federal Acquisition Regulations dealing with various types of federal information or information generated by contractors that the federal deems sensitive. This sensitivity may relate to privacy concerns, national security, or economic competitiveness (FAR, 2021).

The Defense Federal Acquisition Regulations describe cybersecurity requirements that defense contractors are required to follow (Acquisition.gov, 2024). These regulations are based on NIST SP 800-171 (NIST 2024) and are often a source of contention between university and industry when discussing terms of a subcontract from a defense contractor to a university. The university may find itself unable or unwilling to comply with DFARS regulations when they are not applicable to research being conducted by the university.

16.3.5 *Data Consortia Repositories*

Data consortia, data repositories, and registries offer options for depositing, accessing, and sharing data within defined parameters and often within groups of colleagues who share common interests and expertise. Some data consortia are created to share data among entities that are performing research in a specific area. Others are not area or discipline specific. NIH, National Science Foundation (NSF), other federal funding agencies, as well as other funding entities, require inclusion of a data sharing and management plan in research proposals.

The International Committee of Medical Journal Editors published two requirements, effective 2018, that are applicable to journals that want to be listed in ICMJE’s website list of publications: (1) medical journal editors must require registration of clinical trials in a public trials registry at or before the time of first patient enrollment as a condition of consideration for publication, and (2) manuscripts submitted to ICMJE journals that report the results of clinical trials must contain a data-sharing statement as described on ICMJE’s website (ICMJE, 2017). In response to these and other data-sharing requirements, some data-sharing consortia have developed websites to facilitate data sharing for researchers with shared interests, for example:

- *Rare Disease Clinical Research Network* is NIH and member funded, and facilitates collaboration among clinical studies, including study enrollment and data sharing.

- *Inter-University Consortium for Political and Social Research* is an academic consortium for data access and curation in the social and behavioral sciences.
- *National Cancer Institute's Epidemiology and Genomics Research Program* highlights four consortia and describes data-sharing expectations.
- *NSF's National Center for Science and Engineering Statistics* provides the DataHub Consortium to link and analyze data from different sources in new ways.

Other sites list repositories where data can be deposited, even if not part of a consortium:

- The *NIH National Library of Medicine Dataset Catalog* (currently in beta) lists data currently curated for inclusion in their Dataset Catalog as well as dataset metadata from the repositories.

Harvard University Dataverse is a “free data repository open to all researchers from any discipline, both inside and outside of the Harvard community.” The repository supports sharing, archival, citation, and exploration among customizable datasets (Harvard, 2024). Some organizations make data available that they have acquired from sources like public records or direct entry by subjects or participants, or that have formed registries of data such as data related to a disease or health condition derived from medical records. Some public data and registries require membership or other formal processes to access the data, for example:

- *National Ecological Observatory Network (NEON)*—NSF-funded portal to access and collect ecological data for understanding of US ecosystems “continental-scale observation facility operated by Battelle and designed to collect long-term open access ecological data and understand how US ecosystems are changing” (NEON, 2024).
- *World Bank Open Data*—“Free and open access to global development data,” including an Open Data Catalog, an analysis and visualization tool, and a Microdata Library (World Bank, 2024).
- *Data.gov*—Over 300,000 datasets “designed to unleash the power of government open data to inform decisions by the public and policymakers, drive innovation and economic activity, achieve agency missions, and strengthen the foundation of an open and transparent government” (GSA, 2009).
- *National Library of Medicine, ClinicalTrials.gov*—National registry of human clinical trials maintained by the National Library of Medicine under NIH.
- *META FORT*—“Facilitates data sharing and the publication of independent research about Facebook’s role in society, with the right privacy protections in place” (META, 2024).

Data access requirements have also raised expectations for how data from different sources can be curated to ensure validity, comparability, and how to use data for decision-making and policy development. A few of these efforts are

- *DataCite*—“A global community ... to ensure that research outputs and resources are openly available and connected so that their reuse can advance knowledge across and between disciplines, now and in the future.” The group also enables

the creation and management of persistent identifiers (PIDs), integrates services to improve research workflows, and facilitates the discovery and reuse of research outputs and resources” (DataCite, 2022).

- *Re3data*—Global registry funded by the German Research Foundation of “research data repositories that covers research data repositories from different academic disciplines. It includes repositories that enable permanent storage of and access to data sets to researchers, funding bodies, publishers, and scholarly institutions” (Re3data, 2012).
- *Data Curation Network*—“A membership organization of institutional and non-profit data repositories whose vision is to advance open research by making data more ethical, reusable, and understandable” (DCN, 2024).
- *UN DESA*—“Helps countries make informed decisions by providing a wealth of information through our publications and databases and through our support for international deliberations at the United Nations General Assembly, Economic and Social Council (ECOSOC), Commissions, Forums and other bodies” (UN DESA, 2023).

In summary, numerous repositories exist for storing, accessing, and sharing research data, to be used as appropriate, depending on the nature and requirements of the research. Multiple sources also exist to learn more about the management of data repositories.

16.3.6 Data-Sharing Resources

Universities are attempting to respond to the increased generation of data as well as the access, privacy, and security concerns that accompany data-sharing requests and requirements. Mechanisms for storing data, such as university-based repositories and procedures for depositing data to third-party repositories, are being developed. The University of Michigan (U-M), for instance, launched the Research Data Stewardship Initiative under its office of sponsored projects “to support U-M researchers in managing and sharing their research data to improve the transparency, rigor and impact of academic research” (University of Michigan, 2024). Other universities, such as Yale, have enlisted their privacy office to help explain options for data sharing with external parties, including links to the Yale Information Security office to access data classification guidelines and a questionnaire for the user of the data to assess the risks posed by the data, to secure the data, and to determine which Yale IT System to use to match the risk classification (Yale, 2024). Here are more examples of other data-sharing resources:

- The University of Virginia tapped the expertise of its library staff to respond to NIH data management and sharing requirements by providing templates, tools, and guidance about, policies, data sharing plans, and repositories (University of Virginia, 2024).

- International Association of Privacy Professionals is a global information privacy community and resource that helps define, promote, and improve the privacy profession globally. IAPPs website offers a vast number of resources and training tools (IAPP, 2024).
- Trade associations, such as Pharmaceutical Research and Manufacturers of America (PHRMA, 2023), provide guidance tailored for their particular industry. A few companies have committed to data sharing and provide a website to describe procedures for obtaining data from them (Merck, 2017).
- The Federal Demonstration Partnership has developed a series of adaptable Data Use agreement templates that are publicly available (FDP, 2024).
- The Council on Governmental Relations, an association of research universities, affiliated medical centers, and independent research institutes, maintains resources on science and security, the NIH data-sharing policy, and keeps abreast of the effect of various data-related regulations, policies, and trends on research universities (COGR, 2023).

In 2016, an article in *Scientific Data* addressed humans' increasing reliance on computational support to deal with the volume, complexity, and speed of data. "The authors intended to provide guidelines to improve the *Findability, Accessibility, Interoperability, and Reuse* (FAIR) of digital assets. The principles emphasize machine-actionability (i.e., the capacity of computational systems to find, access, interoperate, and reuse data with none or minimal human intervention" (Wilkinson et al., 2016). Following publication, a stakeholder-driven self-governed initiative was formed to implement the FAIR principles. The GO FAIR website lists contact information for Data Stewardship Competence Centers' (DSCC) that collaborate with researchers in their institutions to enable better data management through compliance with FAIR principles. The site also includes a research data management starter kit with links to various related resources (GO FAIR, 2024).

16.4 Takeaways and Future Strategies

Data is a complex topic and data sharing even more so. UIDP Contract Accord 14 (Data) reviews many of the key issues from industry and university perspectives, including a list of principles that both parties would generally agree on and terms to be considered in drafting and reviewing data clauses in UI agreements (UIDP, 2020).

Perhaps the key lesson from this chapter is that despite all the laws, rules, policies, lists and resources, discussions between a university and company that involve generation of, access to, or use of data depend on the nature of the interaction, the project, and the objectives and intentions of the parties. Data issues should be considered from at least four key perspectives: privacy, regulation, security, and value. Data should be clearly defined. Related deliverables should be identified and related rights and obligations of the parties should be described with enough detail for the parties to understand and conform to them.

Features of Data Clauses from UIDP Contract Accord 14

1. *A clear description of Data to be provided;*
2. *Permitted uses for the Data and any regulatory requirements that the Provider needs to have in place;*
3. *Names or general descriptions of individuals who can access or receive the Data;*
4. *Conditions under which the User can provide the Data to other Users;*
5. *Length of time the Data is to be made available by the Provider and retained or used by the User;*
6. *Method of Data disposal at the end of the use period (returned or destroyed);*
7. *The User's obligations regarding new Data generated that is based on the Data originally provided;*
8. *Management of new intellectual property created using the Data;*
9. *Instructions on how the Data should be aggregated, encrypted, anonymized, or de-identified;*
10. *Safeguards required to protect confidential, private, and sensitive information;*
11. *Process for review by the Provider of publications resulting from use of the Data;*
12. *Practical aspects of the Data transfer (e.g., where, when, how); and*
13. *Statement of ownership of the Data if it is proprietary, and the provenance and authenticity of the Data if that requires confirmation.*

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